Leuven Statistics Day 2023

Reliability for multilevel data: A correlation approach

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- Two recent papers (ten Hove et al., 2022; Schönbrodt et al., 2022) address how to quantify reliability in multilevel settings based on generalizability theory.
  - Nevertheless, both approaches give **inconsistent results** regarding their definition of cluster-level reliability.

Classical Test Theory

An observed-score Y_{ij} for subject i at occasion j can be decomposed into two parts: a true score T_{ij} and a random measurement error ϵ_{ij}

$$Y_{ij} = T_{ij} + \epsilon_{ij}$$

- where $E[Y_{ij}] = T_{ij}, Cor(\epsilon_{ij}, \epsilon_{ij'}) = Cor(\epsilon_{ij}, T_{ij}) = 0$

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The reliability is defined as

$$Cor(Y_{ij}, T_{ij})^2 = \frac{\sigma_T^2}{\sigma_Y^2} = \frac{\sigma_T^2}{\sigma_T^2 + \sigma_\epsilon^2}$$

- It is a ratio of the true-score variance σ_T^2 over the total variance σ_Y^2.

Classical Test Theory

Under the parallel assumptions

- $E[Y_{ij}] = E[Y_{ij'}] = T_i$, $Var(Y_{ij}) = Var(Y_{ij'})$,
- The reliability coefficient is also equal to the correlation between two replicates Y_{ij} and $Y_{ij'}$, i.e.,

$$Cor(Y_{ij}, Y_{ij'}) = \frac{\sigma_T^2}{\sigma_T^2 + \sigma_\epsilon^2} = Cor(Y_{ij}, T_{ij})^2.$$

Generalizability Theory

Extending classical test theory

- Recognize and estimate the magnitude of the multiple sources of variation (Cronbach, 1972; Brennan, 2001).
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- Recognize and estimate the magnitude of the multiple sources of variation (Cronbach, 1972; Brennan, 2001).
  - such as subjects, raters, items, occasions, etc.
- E.g., a three-way $Subject(S_i) \times Rater(R_j) \times Day(D_t)$ design

	Rater 1		Rater 2	
Subject	Day 1	Day 7	Day 1	Day 7
1	Y_{111}	Y_{112}	Y_{121}	Y_{122}
2	Y_{211}	Y_{212}	Y_{221}	Y_{222}
3	Y_{311}	Y_{312}	Y_{321}	Y_{322}
.	.	.	.	.
.	.	.	.	.
10	Y_{1011}	Y_{1012}	Y_{1021}	Y_{1022}

$$Y_{ijt} = \mu + S_i + R_j + D_t + SR_{ij} + SD_{it} + RD_{jt} + \epsilon_{ijt}$$

GT: $Y_{ijt} = \mu + S_i + R_j + D_t + SR_{ij} + SD_{it} + RD_{jt} + \epsilon_{ijt}$

Relative reliability (R_{rel})

- Consistency between subjects' rank-orderings

$$\frac{\sigma_S^2}{\sigma_S^2 + \sigma_{SR}^2 + \sigma_{SD}^2 + \sigma_\epsilon^2}$$

Absolute reliability (R_{abs})

- Agreement between subjects' absolute scores

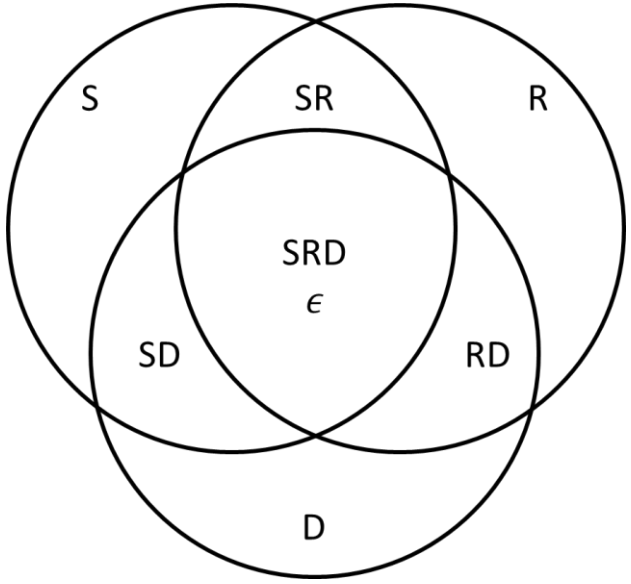
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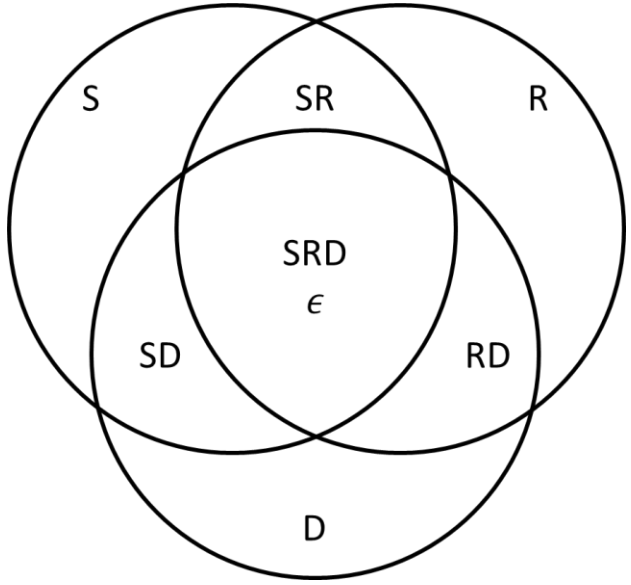
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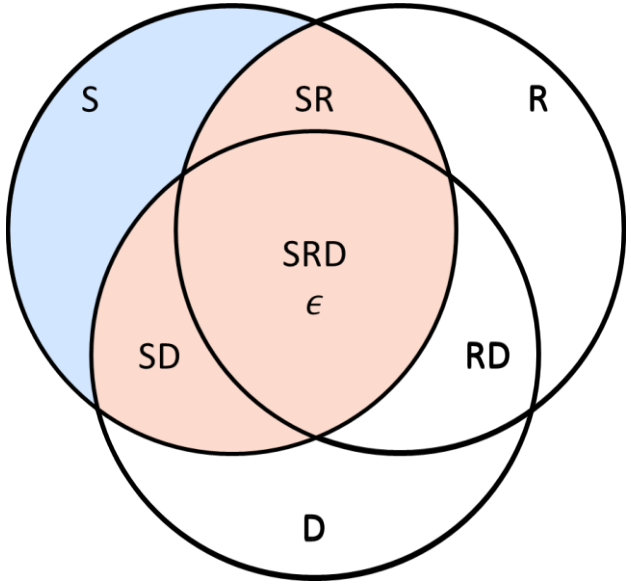


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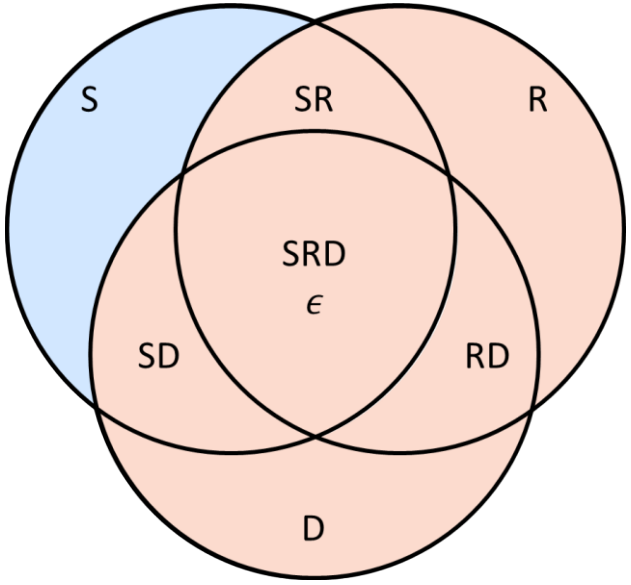
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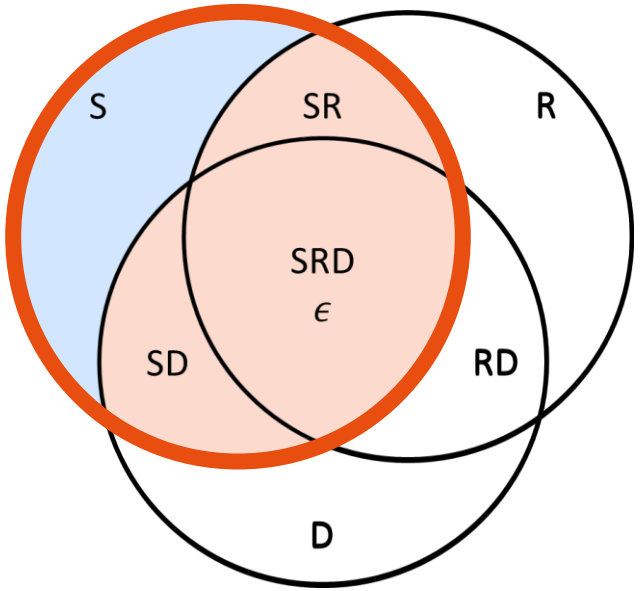


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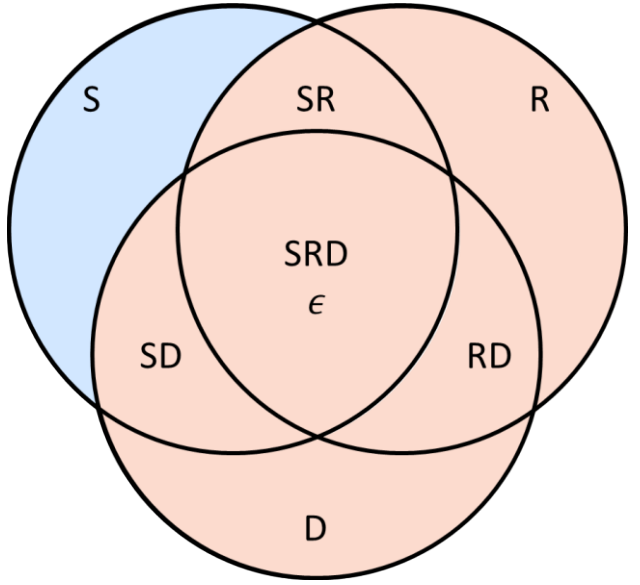
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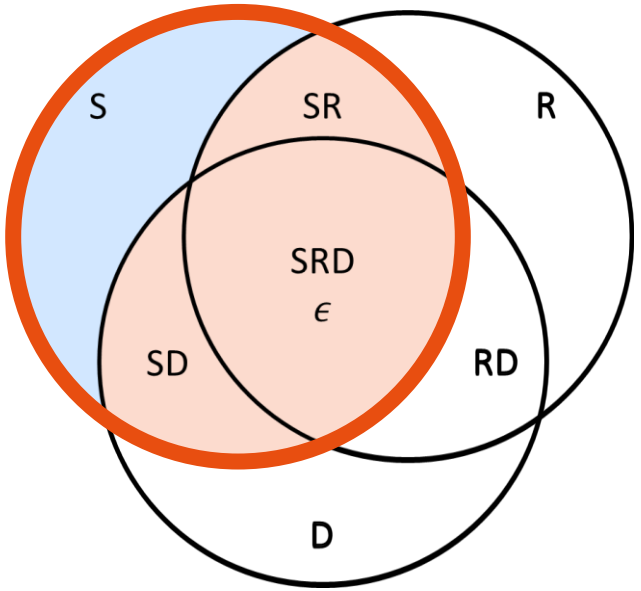


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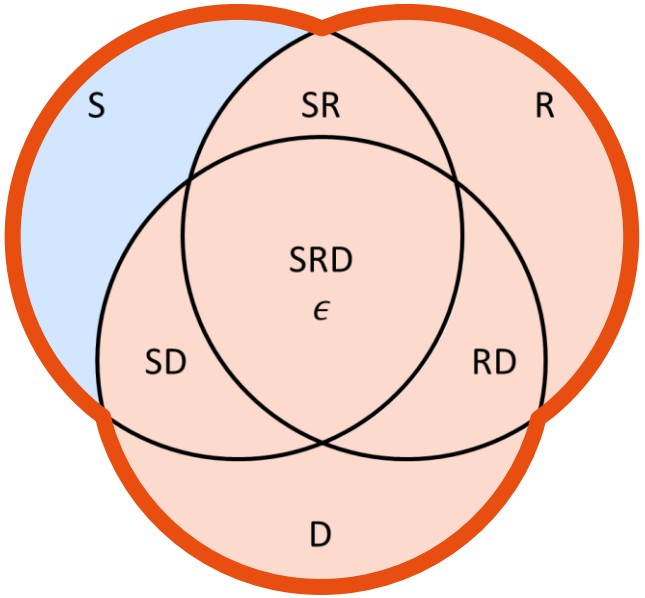
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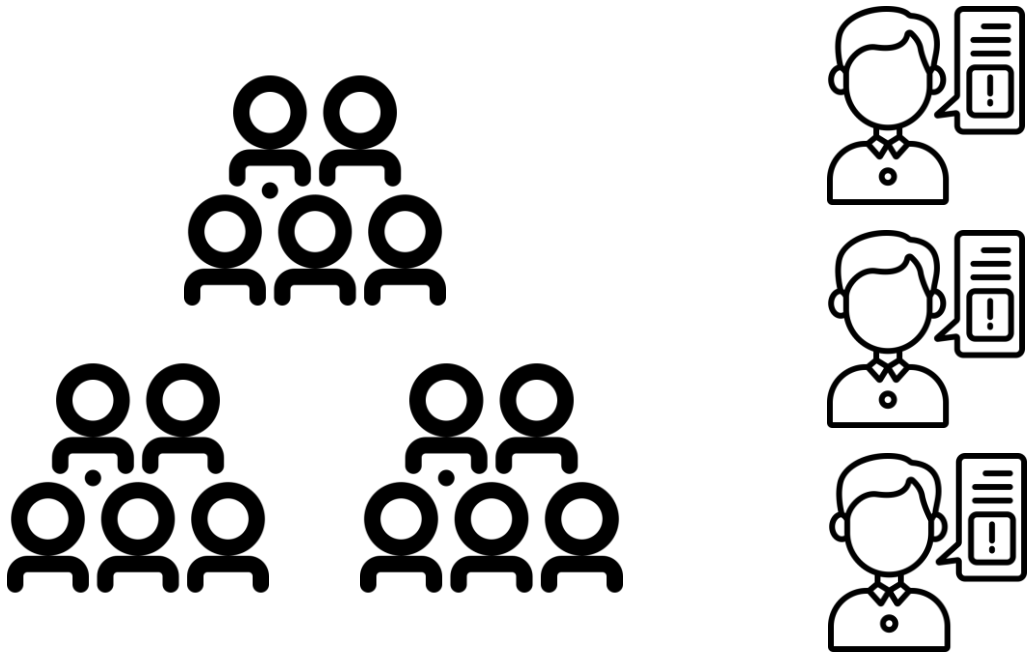
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Nested Structures

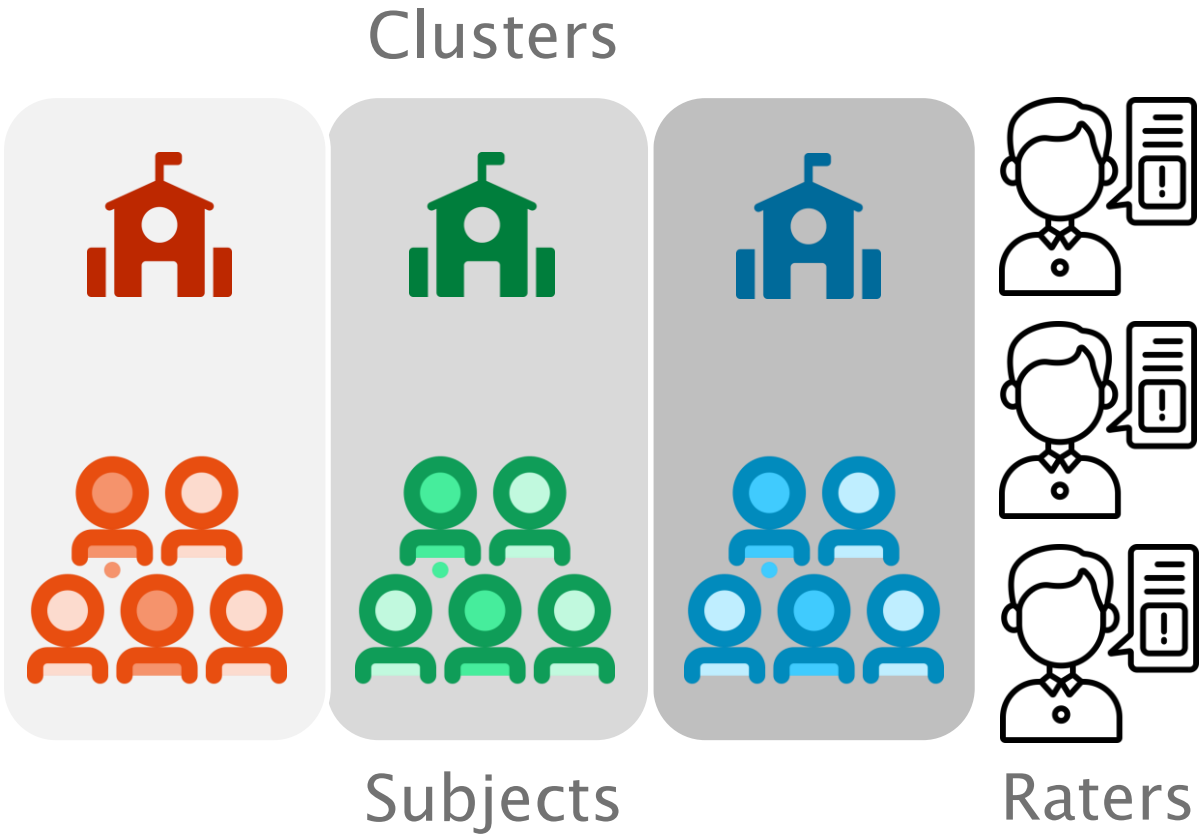
E.g., subjects ($S_{i(k)}$) are nested within clusters (C_k), and raters (R_j) are crossed with S & C.



Subjects Raters

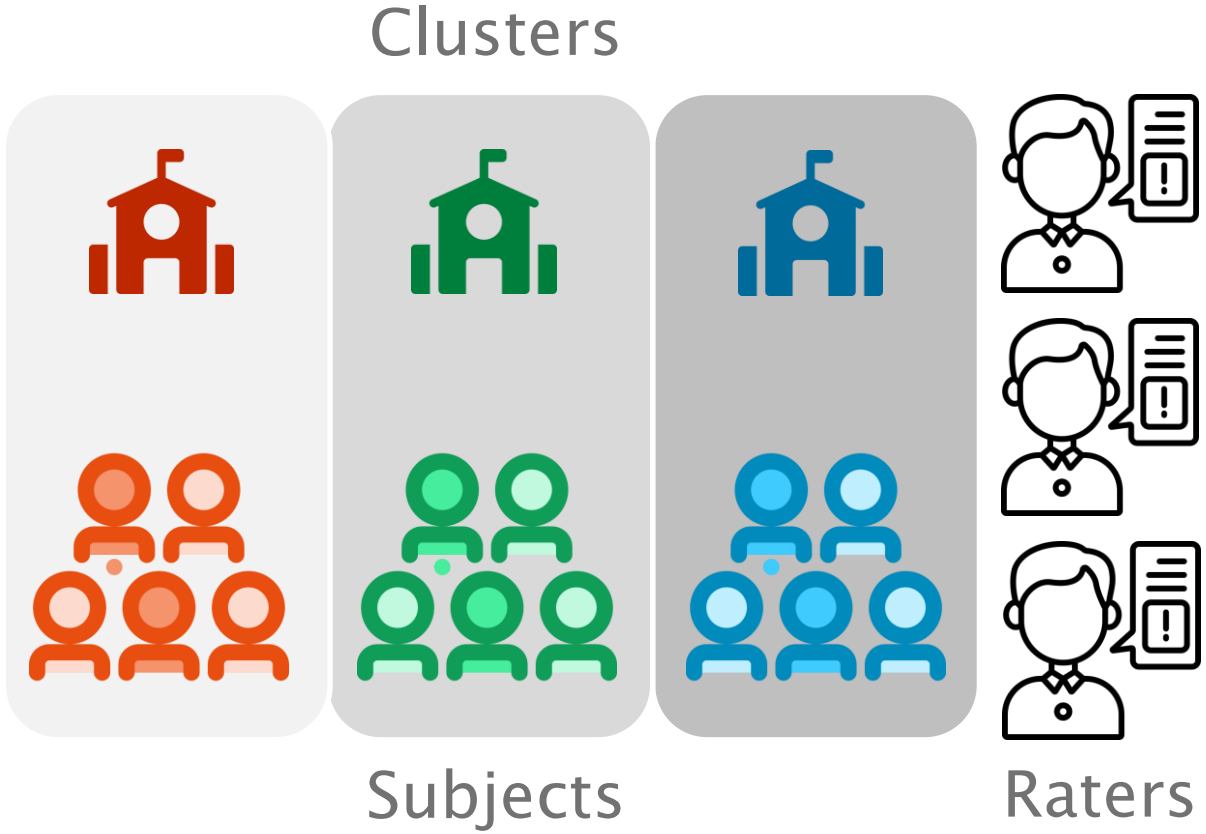
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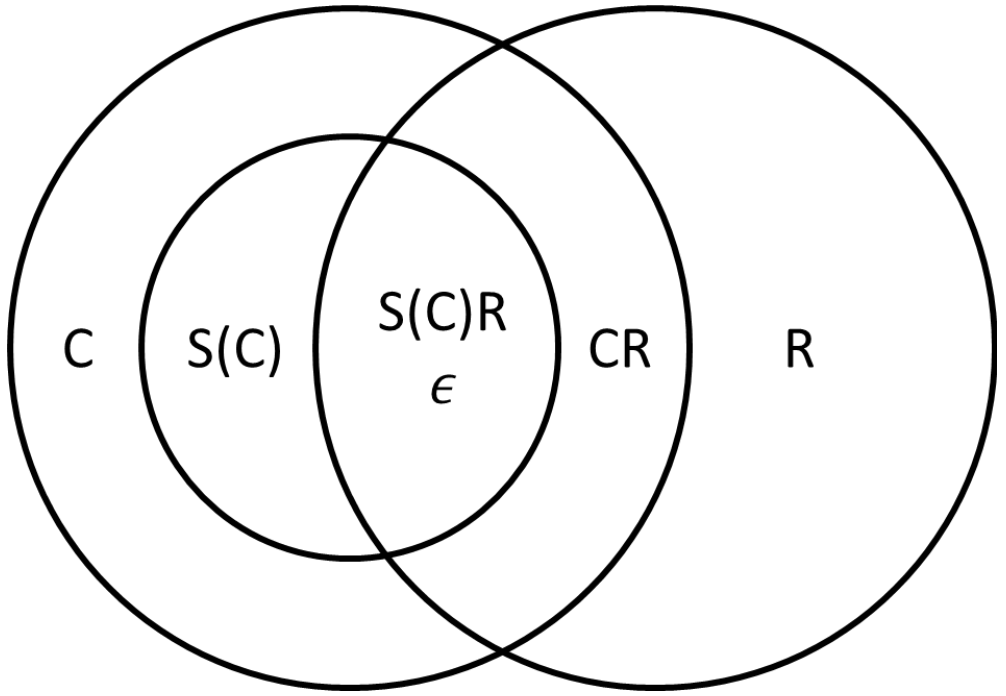


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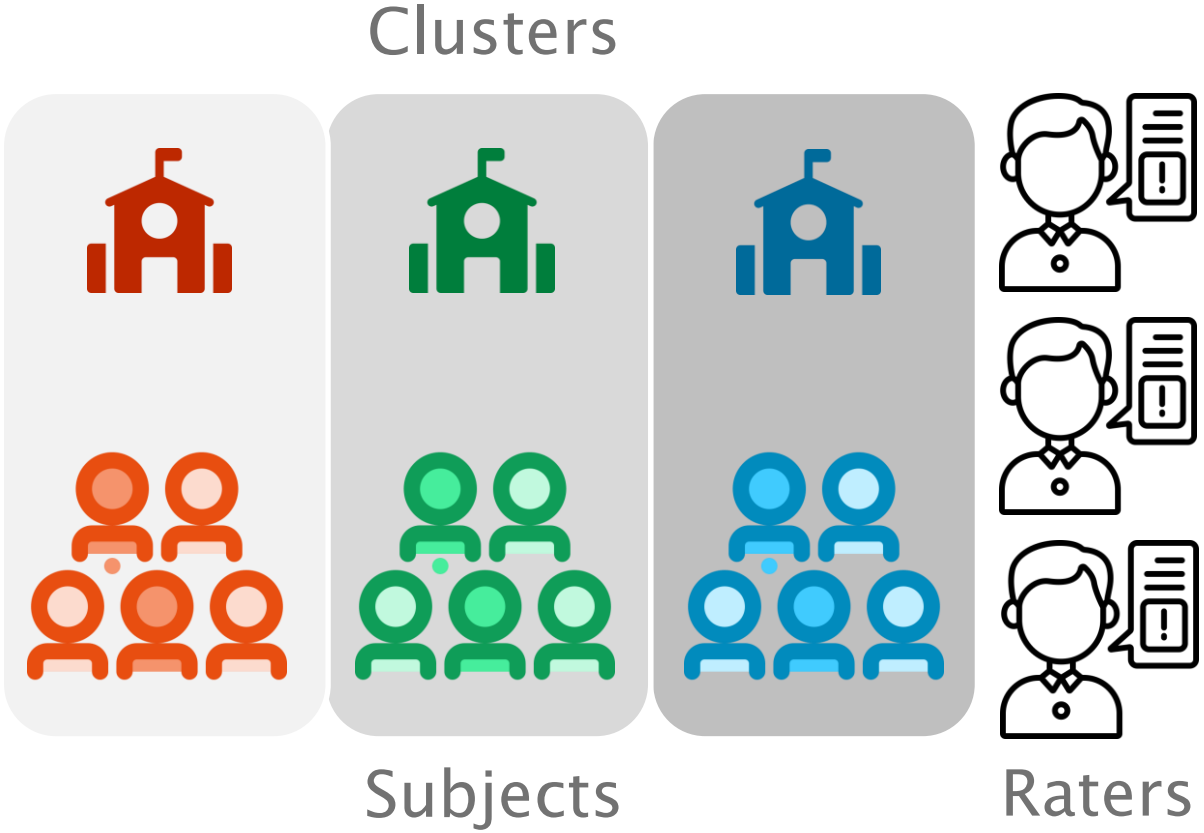


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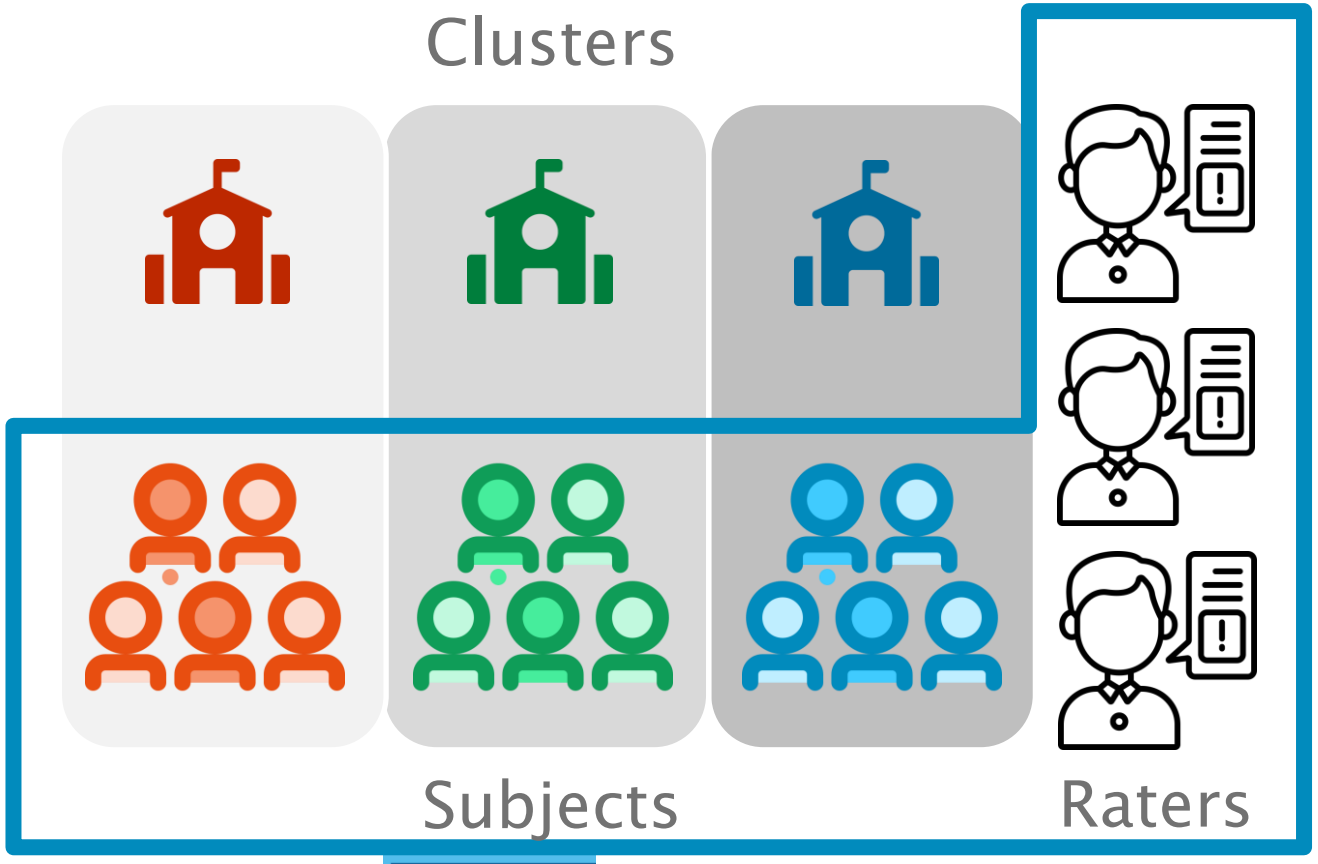
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Reliability for the different levels?
Subject / Cluster

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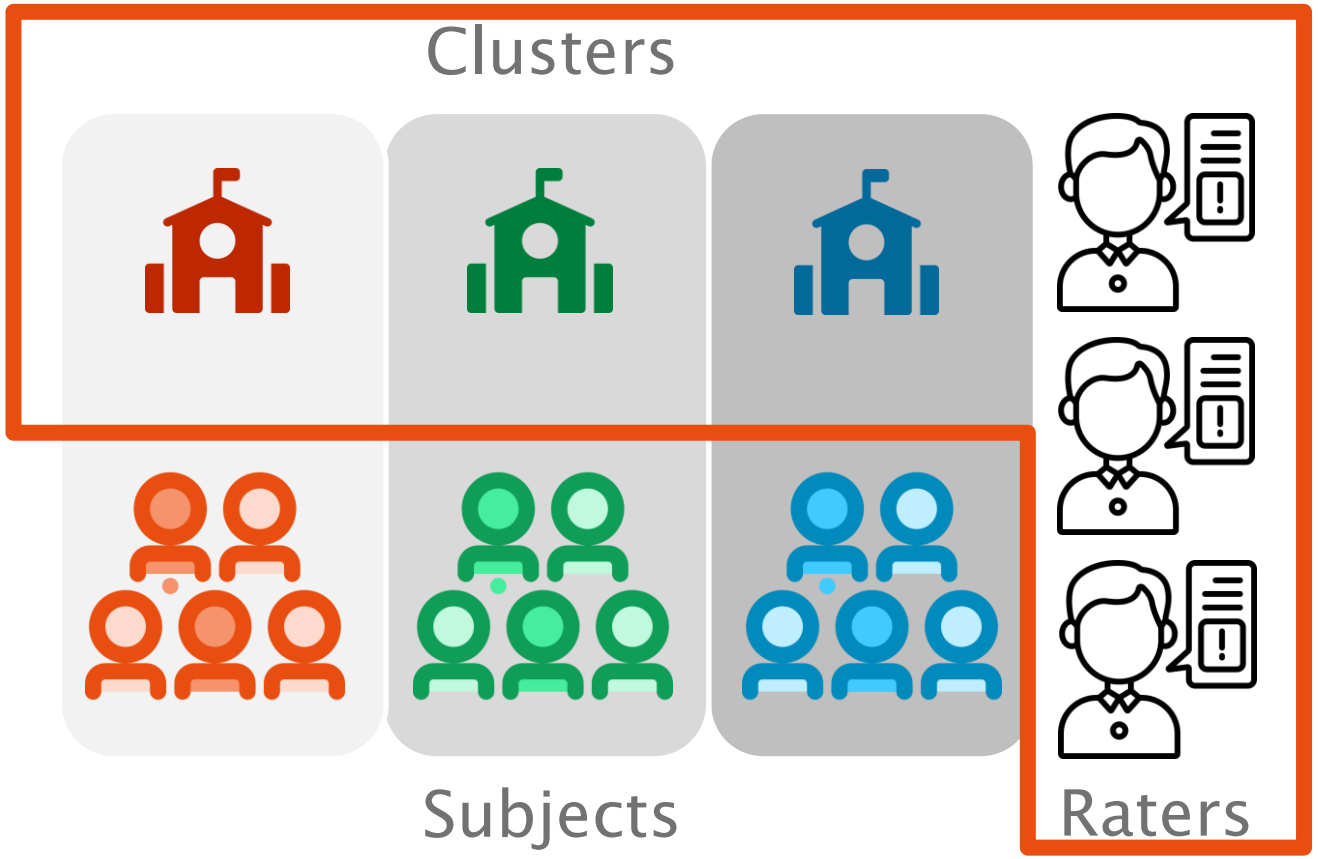
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Interrater Reliability for Multilevel Data: A Generalizability Theory Approach

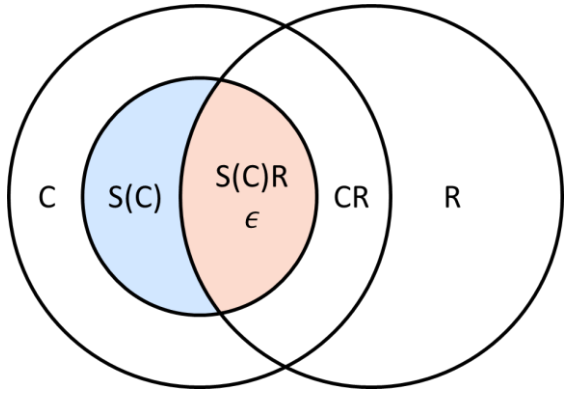
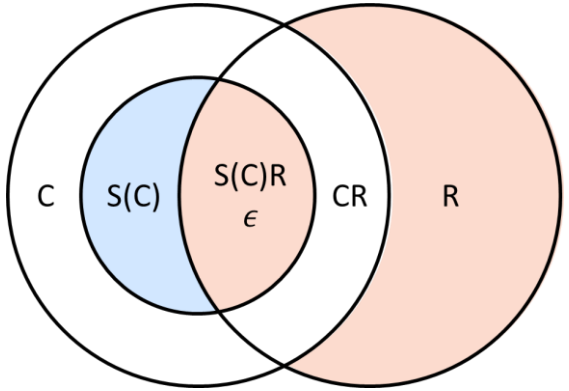
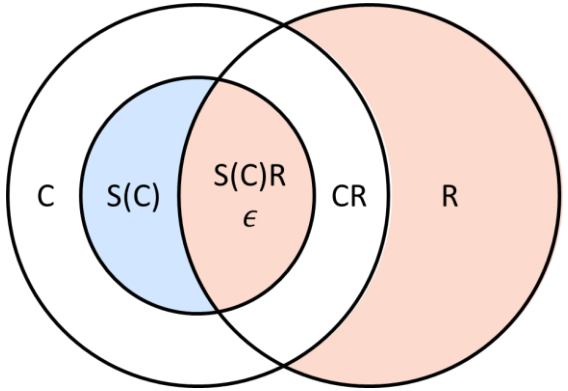
Debby ten Hove, Terrence D. Jorgensen, and L. Andries van der Ark
Research Institute of Child Development and Education, University of Amsterdam

Measuring motivational relationship processes in experience sampling: A reliability model for moments, days, and persons nested in couples

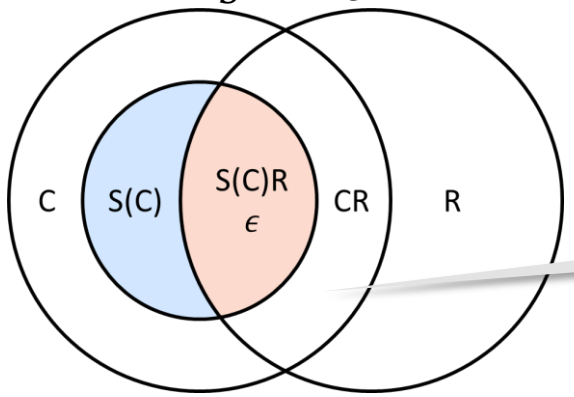
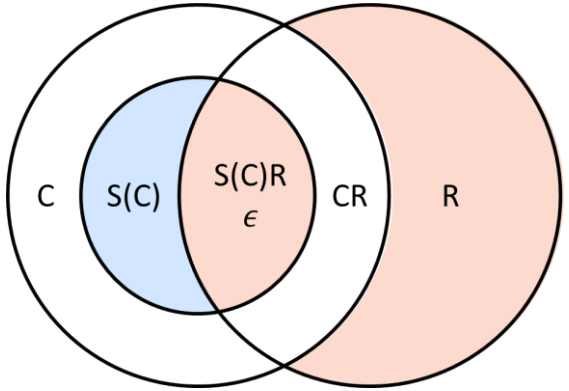
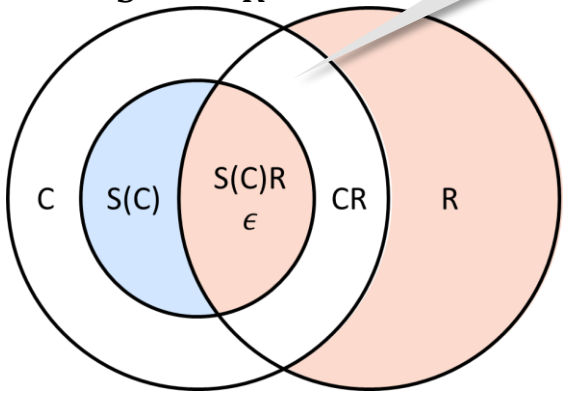
Behavior Research Methods (2022) 54:1869–1888
https://doi.org/10.3758/s13428-021-01701-7

Felix D. Schönbrodt[1] · Caroline Zygar-Hoffmann[1] · Steffen Nestler[2] · Sebastian Pusch[3] · Birk Hagemeyer[3]

Case 1: $Y_{i(k)j} = \mu + C_k + S_{i(k)} + R_j + CR_{kj} + \epsilon_{i(k)j}$

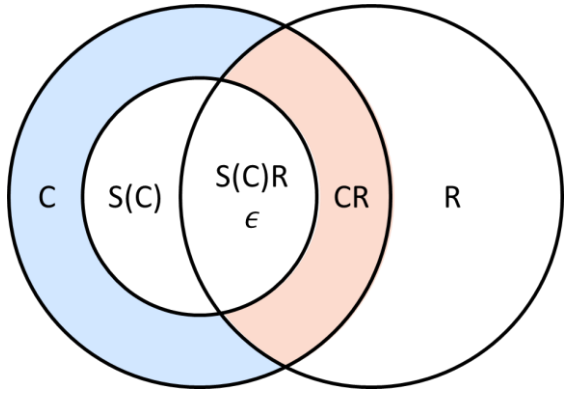
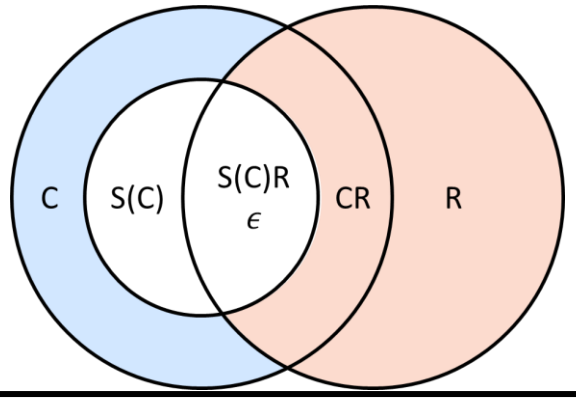
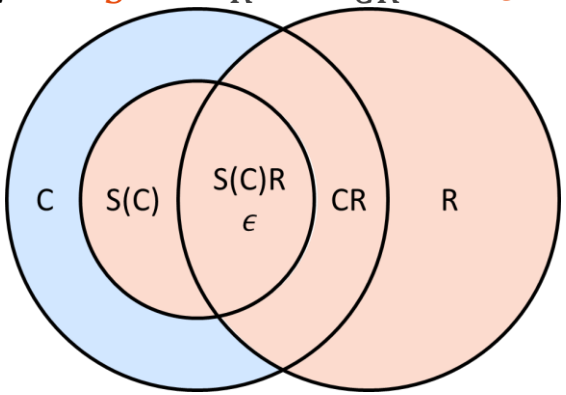
	ten Hove et al. (2022)	Schönbrodt et al. (2022)
Subject-level	$\frac{\sigma_S^2}{\sigma_S^2 + \sigma_\epsilon^2}$	
R^S_{rel}		(not defined)
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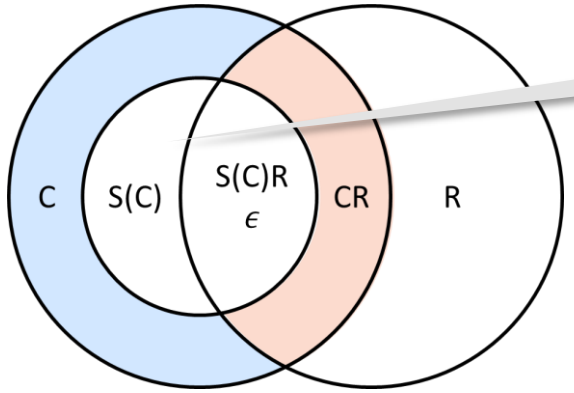
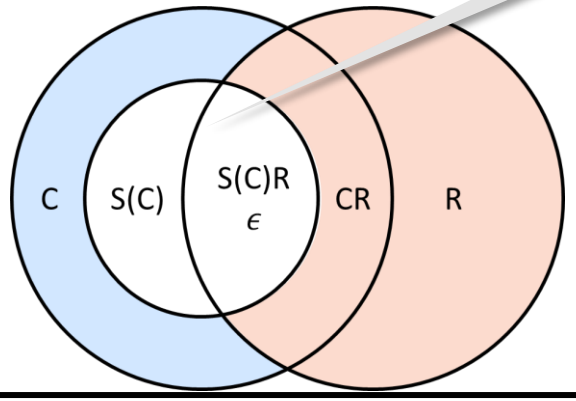
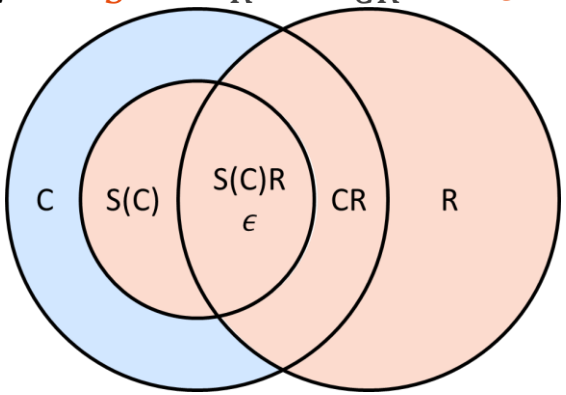
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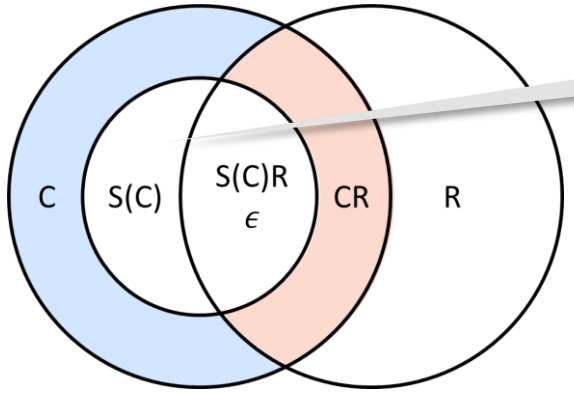
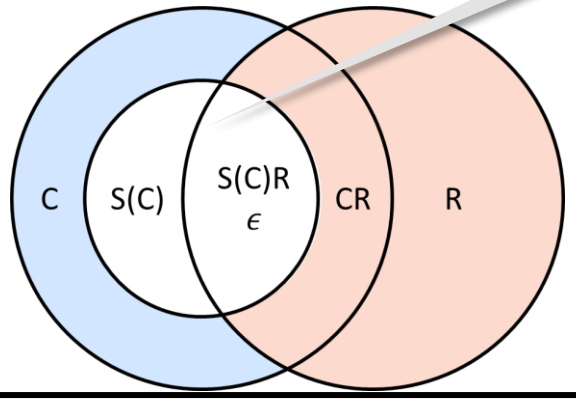
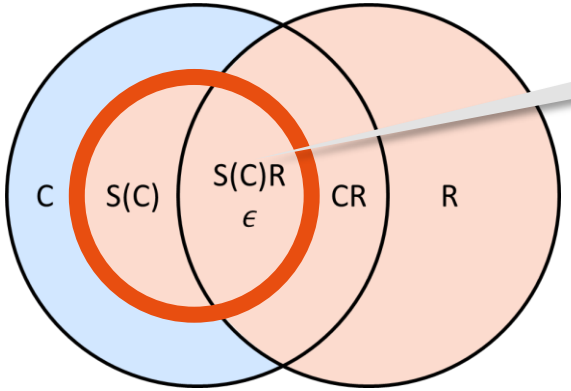
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Ignore subject-level components

Include everything

Comparison

	ten Hove et al. (2022)	Schönbrodt et al. (2022)
Subject level	Do not contain any cluster-related (higher level) variance components (i.e., σ_C^2, σ_{CR}^2)	
Cluster level	Do not contain any subject-related variance (lower level) components (i.e., $\sigma_{S(C)}^2, \sigma_{S(C)R}^2$)	Contain subject-related variance components
Types	Separate **relative** and **absolute** reliability	Only discuss **absolute** reliability

Correlation Approach for Multilevel Data

The reliability coefficient can be defined as the **correlation** between two measurements from the same subject i on different raters (or occasions, etc.) j and j' (Vangeneugden et al., 2005).

	Subject-level	Cluster-level
Relative reliability	$\mathrm{Cor}\left(Y_{i(k)j}, Y_{i(k)j'} \mid k, j, j'\right)$	$\mathrm{Cor}\left(Y_{i(k)j}, Y_{i(k)j'} \mid i(k), j, j'\right)$
Absolute reliability	$\mathrm{Cor}\left(Y_{i(k)j}, Y_{i(k)j'} \mid k\right)$	$\mathrm{Cor}\left(Y_{i(k)j}, Y_{i(k)j'} \mid i(k)\right)$

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The reliability coefficient can be defined as the **correlation** between two measurements from the same subject i on different raters (or occasions, etc.) j and j' (Vangeneugden et al., 2005).

	Subject-level	Cluster-level
Relative reliability	$\mathrm{Cor}\left(Y_{i(k)j}, Y_{i(k)j'} \mid k, j, j' \right)$	$\mathrm{Cor}\left(Y_{i(k)j}, Y_{i(k)j'} \mid i(k), j, j' \right)$
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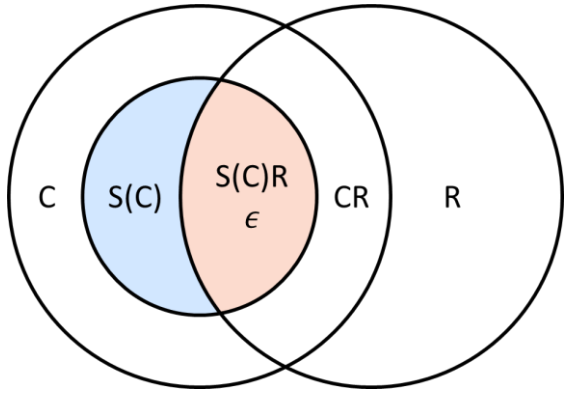
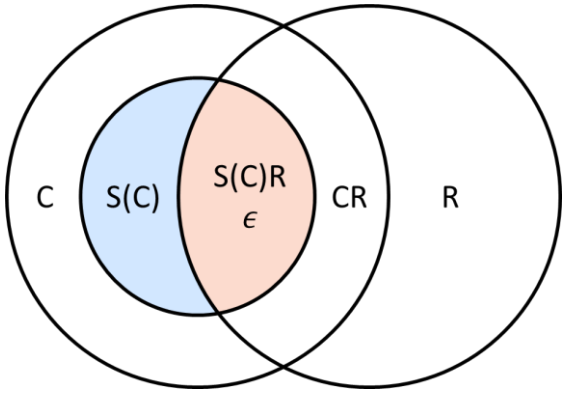
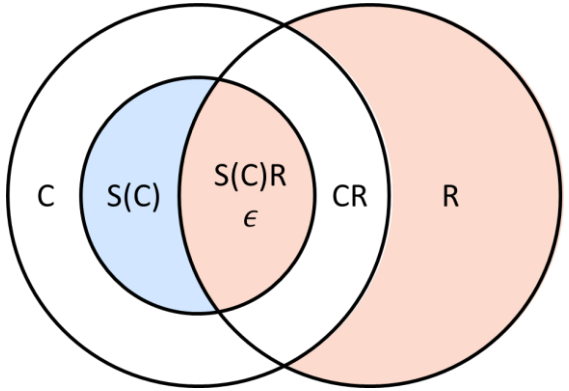
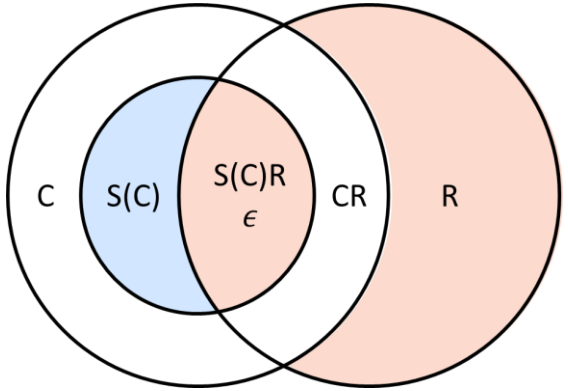
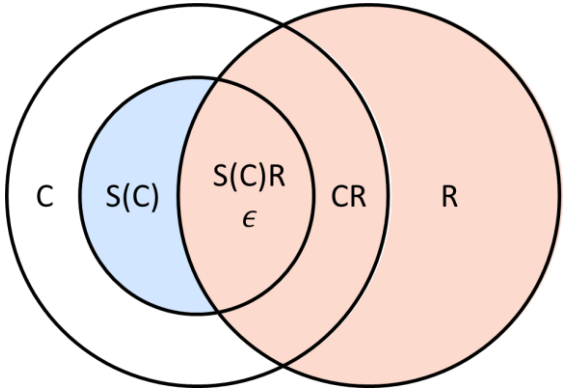
Correlation Approach for Multilevel Data

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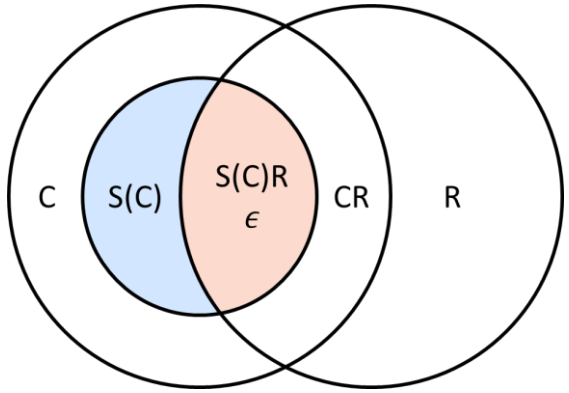
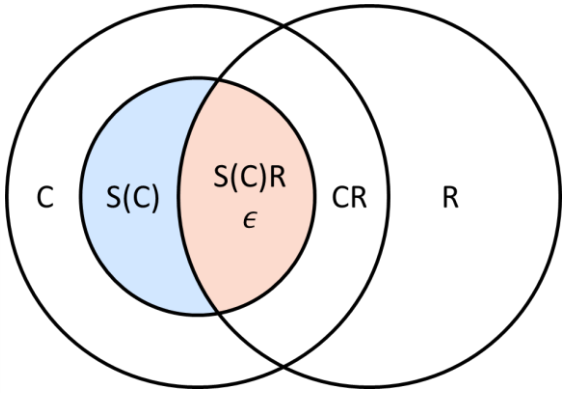
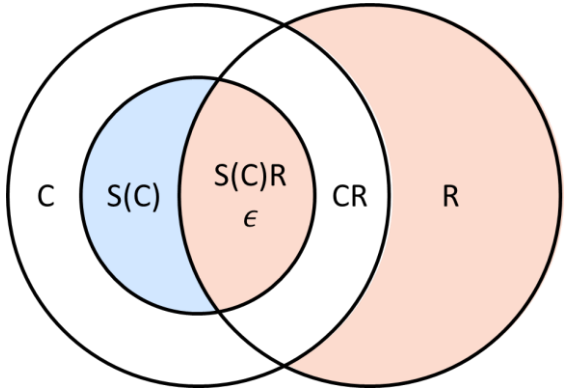
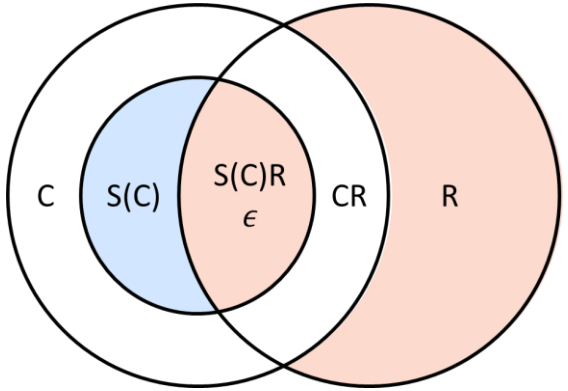
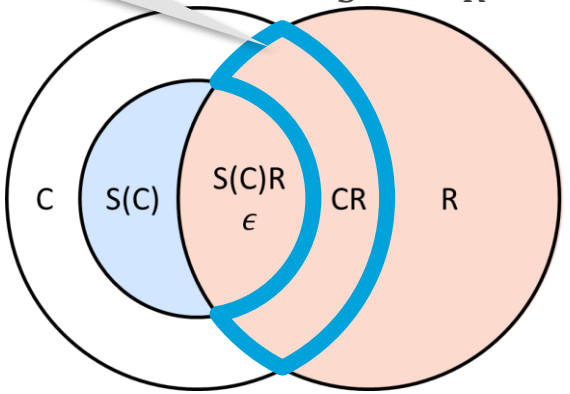
	Subject-level	Cluster-level
Relative reliability	$\text{Cor}(Y_{i(k)j}, Y_{i(k)j'} \mid k, j, j')$	$\text{Cor}(Y_{i(k)j}, Y_{i(k)j'} \mid i(k), j, j')$
Absolute reliability	$\text{Cor}(Y_{i(k)j}, Y_{i(k)j'} \mid k)$	$\text{Cor}(Y_{i(k)j}, Y_{i(k)j'} \mid i(k))$

$C_k, R_j, R_{j'}, CR_{kj}, CR_{kj'}$

Case 1: $Y_{i(k)j} = \mu + C_k + S_{i(k)} + R_j + CR_{kj} + \epsilon_{i(k)j}$

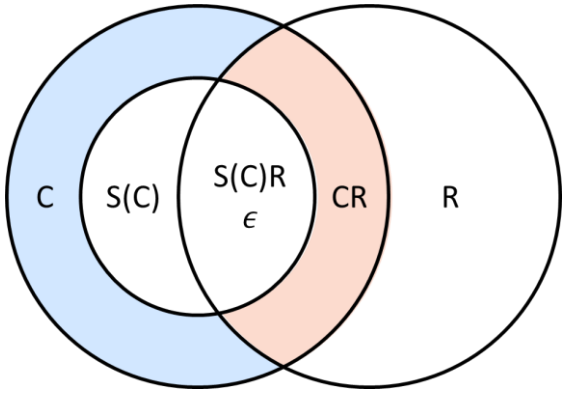
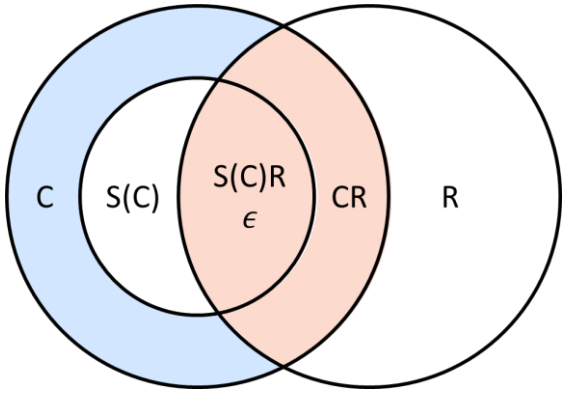
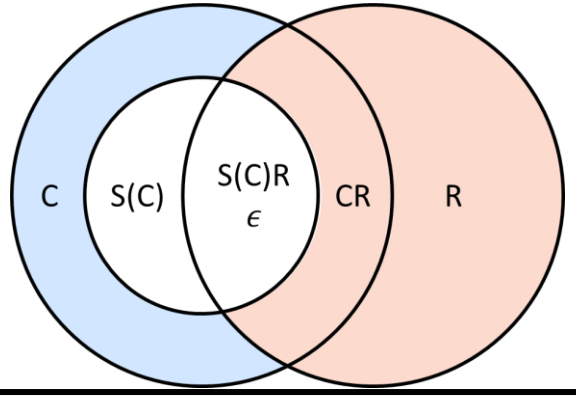
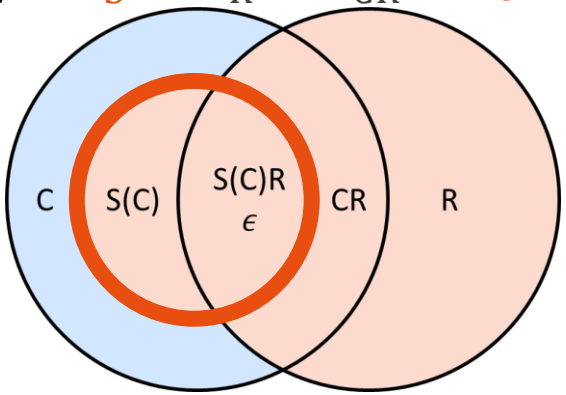
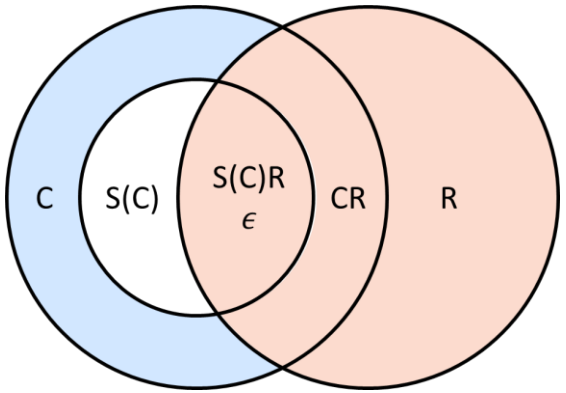
	ten Hove et al. (2022)	Schönbrodt et al. (2022)	Correlation approach
Subject-level	$\frac{\sigma_S^2}{\sigma_S^2 + \sigma_\epsilon^2}$		$\mathrm{Cor}\left(Y_{i(k)j}, Y_{i(k)j'} \mid k, j, j' \right) = \frac{\sigma_S^2}{\sigma_S^2 + \sigma_\epsilon^2}$
R^S_{rel}		(not defined)	
	$\frac{\sigma_S^2}{\sigma_S^2 + \sigma_R^2 + \sigma_\epsilon^2}$	$\frac{\sigma_S^2}{\sigma_S^2 + \sigma_R^2 + \sigma_\epsilon^2}$	$\mathrm{Cor}\left(Y_{i(k)j}, Y_{i(k)j'} \mid k \right) = \frac{\sigma_S^2}{\sigma_S^2 + \sigma_R^2 + \sigma_{CR}^2 + \sigma_\epsilon^2}$
R^S_{abs}			

Case 1: $Y_{i(k)j} = \mu + C_k + S_{i(k)} + R_j + CR_{kj} + \epsilon_{i(k)j}$

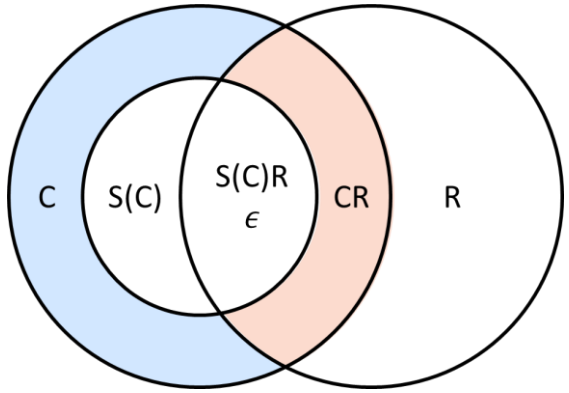
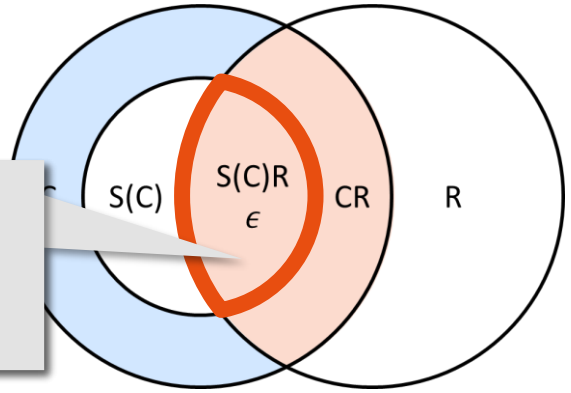
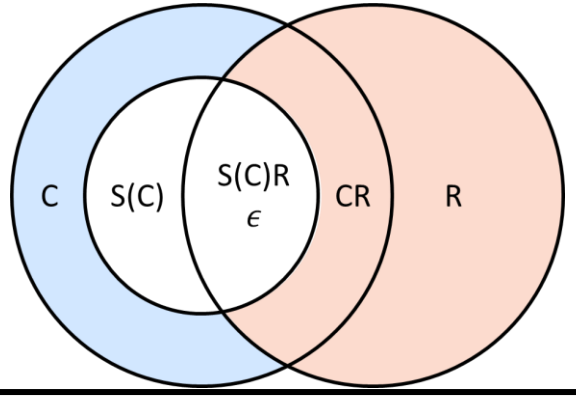
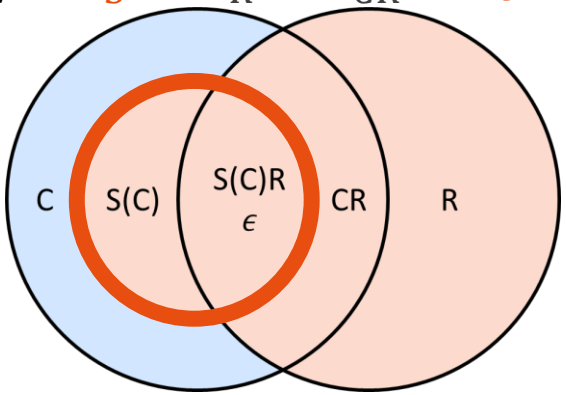
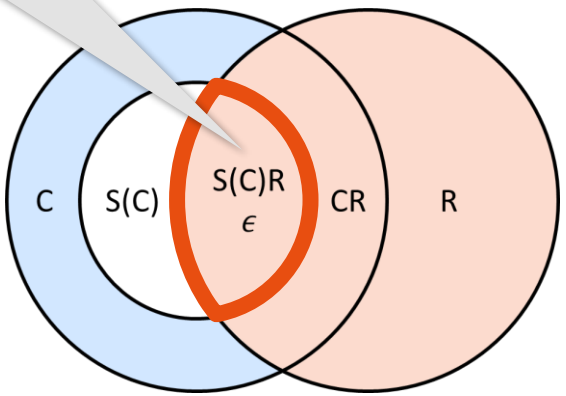
	ten Hove et al. (2022)	Schönbrodt et al. (2022)	Correlation approach
Subject-level	$\frac{\sigma_S^2}{\sigma_S^2 + \sigma_\epsilon^2}$	(not defined)	$\mathrm{Cor}\left(Y_{i(k)j}, Y_{i(k)j'} \mid k, j, j'\right) = \frac{\sigma_S^2}{\sigma_S^2 + \sigma_\epsilon^2}$
R^S_{rel}			
R^S_{abs}	$\frac{\sigma_S^2}{\sigma_S^2 + \sigma_R^2 + \sigma_\epsilon^2}$ 	$\frac{\sigma_S^2}{\sigma_S^2 + \sigma_R^2 + \sigma_\epsilon^2}$ 	$\mathrm{Cor}\left(Y_{i(k)j}, Y_{i(k)j'} \mid k\right) = \frac{\sigma_S^2}{\sigma_S^2 + \sigma_R^2 + \sigma_{CR}^2 + \sigma_\epsilon^2}$ 

Because CR is random and influences rater effects

Case 1: $Y_{i(k)j} = \mu + C_k + S_{i(k)} + R_j + CR_{kj} + \epsilon_{i(k)j}$

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Cluster-level	$\frac{\sigma_C^2}{\sigma_C^2 + \sigma_{CR}^2}$	(not defined)	$\mathrm{Cor}\left(Y_{i(k)j}, Y_{i(k)j'} \mid i(k), j, j' \right) = \frac{\sigma_C^2}{\sigma_C^2 + \sigma_{CR}^2 + \sigma_\epsilon^2}$
R^2_{rel}			
	$\frac{\sigma_C^2}{\sigma_C^2 + \sigma_R^2 + \sigma_{CR}^2}$	$\frac{\sigma_C^2}{\sigma_C^2 + \sigma_S^2 + \sigma_R^2 + \sigma_{CR}^2 + \sigma_\epsilon^2}$	$\mathrm{Cor}\left(Y_{i(k)j}, Y_{i(k)j'} \mid i(k) \right) = \frac{\sigma_C^2}{\sigma_C^2 + \sigma_R^2 + \sigma_{CR}^2 + \sigma_\epsilon^2}$
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	ten Hove et al. (2022)	Schönbrodt et al. (2022)	Correlation approach
Cluster-level	$\frac{\sigma_C^2}{\sigma_C^2 + \sigma_{CR}^2}$	(not defined)	$\mathrm{Cor}\left(Y_{i(k)j}, Y_{i(k)j'} \mid i(k), j, j' \right) = \frac{\sigma_C^2}{\sigma_C^2 + \sigma_{CR}^2 + \sigma_\epsilon^2}$
R^C_{rel}			
	$\frac{\sigma_C^2}{\sigma_C^2 + \sigma_R^2 + \sigma_{CR}^2}$	$\frac{\sigma_C^2}{\sigma_C^2 + \sigma_S^2 + \sigma_R^2 + \sigma_{CR}^2 + \sigma_\epsilon^2}$	$\mathrm{Cor}\left(Y_{i(k)j}, Y_{i(k)j'} \mid i(k) \right) = \frac{\sigma_C^2}{\sigma_C^2 + \sigma_R^2 + \sigma_{CR}^2 + \sigma_\epsilon^2}$
R^C_{abs}			

With measurement error variance

Discussion and Future Work

- Point out the different reliability definitions for multilevel data in the latest literature (ten Hove et al., 2022; Schönbrodt et al., 2022).
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- Challenges
  - reliability coefficients for linear mixed-effects models are inconsistencies between ICC, GT, and the correlation approach
  - the meaning of conditioning

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Q&A

Reliability for multilevel data: A correlation approach

Tzu-Yao Lin, Francis Tuerlinckx, Sophie Vanbelle

Reliability for other nested structures lists in the appendix slides

Maastricht University KU LEUVEN

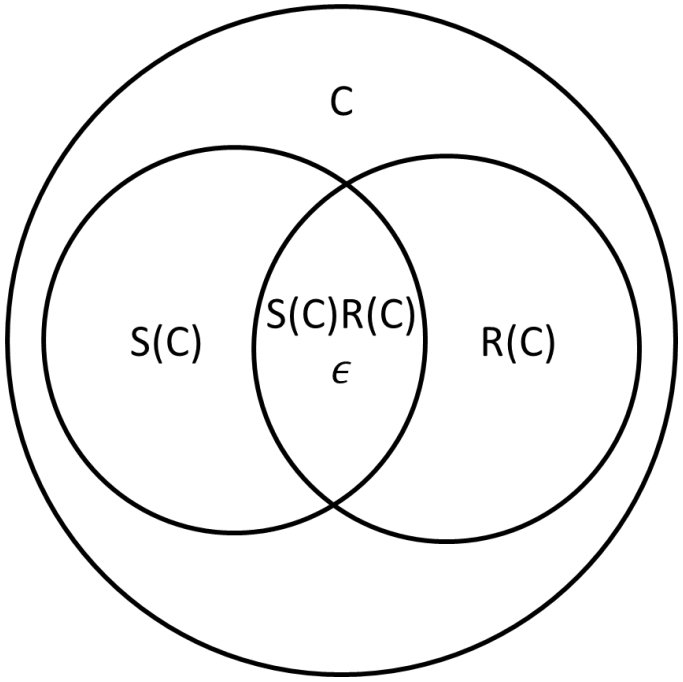
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Other Nested Structure Examples

Subjects (S) are nested within clusters (C), and …

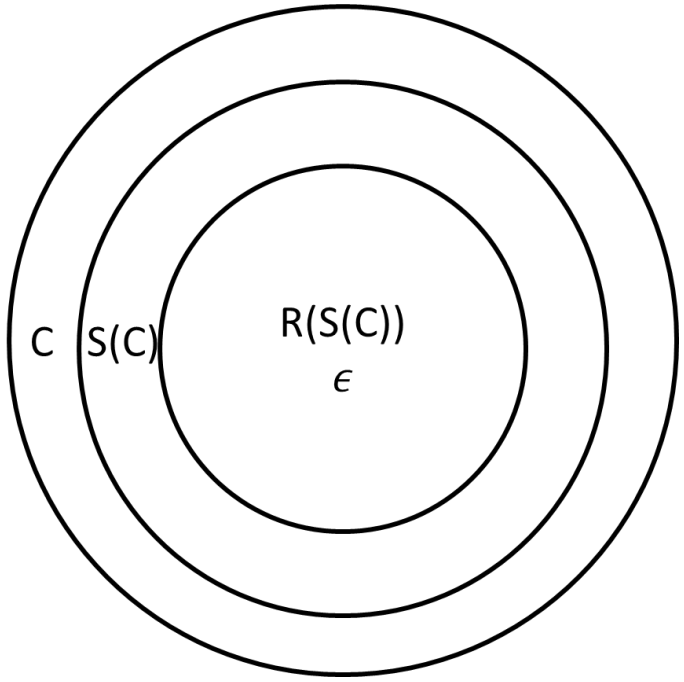
- Case 2: raters (R) are *nested within* C and *crossed with* S

$$Y_{i(k)j(k)} = \mu + C_k + S_{i(k)} + R_{j(k)} + \epsilon_{i(k)j(k)}$$

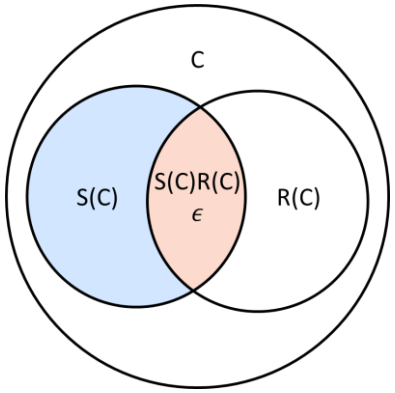
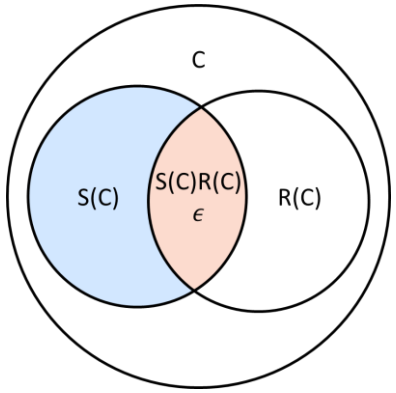
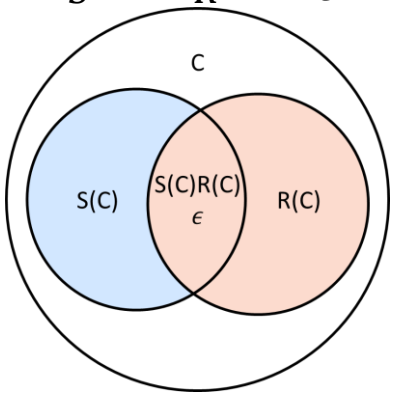
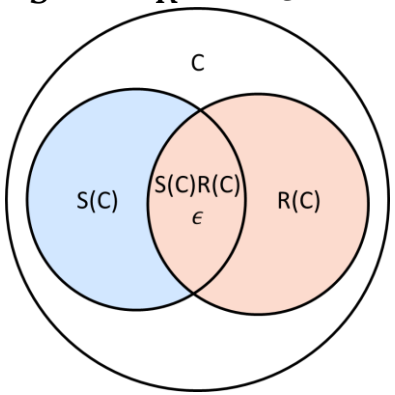
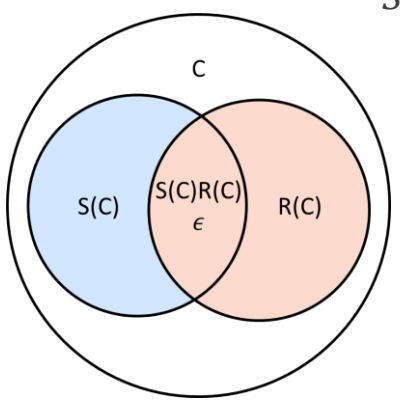


- Case 3: raters (R) are *nested within* S & C

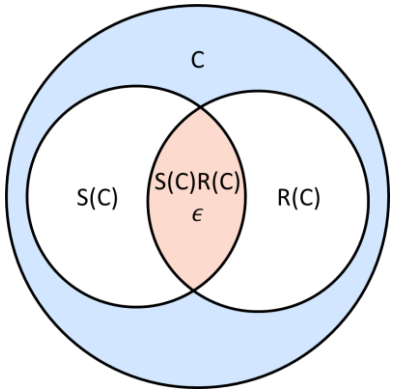
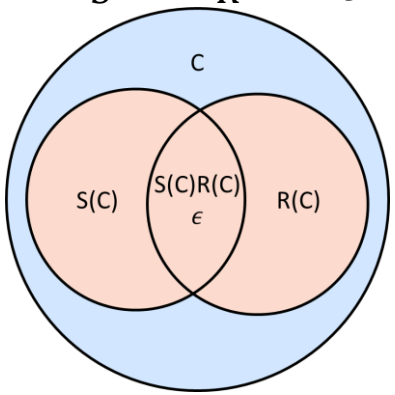
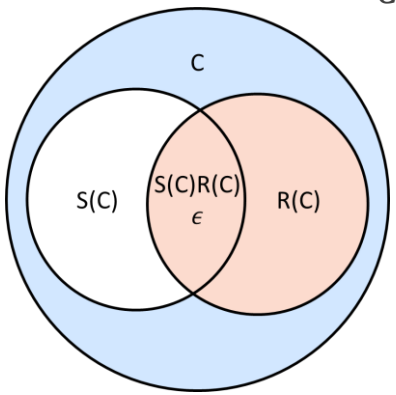
$$Y_{j(i(k))} = \mu + C_k + S_{i(k)} + \epsilon_{j(i(k))}$$



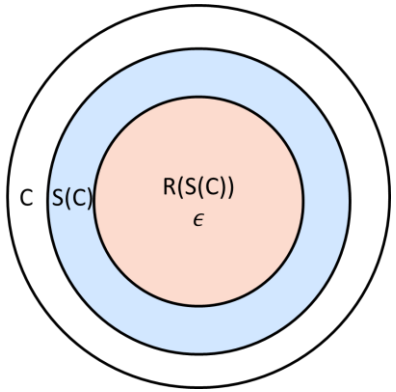
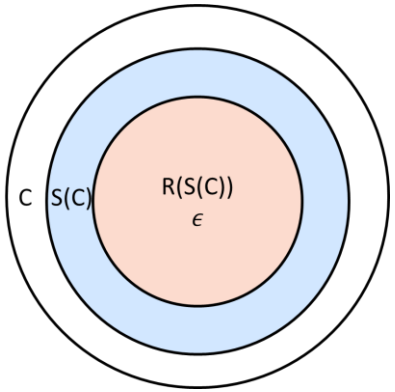
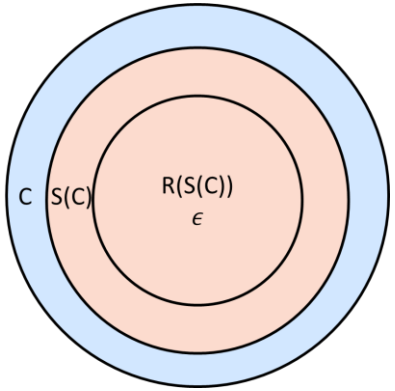
Case 2: $Y_{i(k)j(k)} = \mu + C_k + S_{i(k)} + R_{j(k)} + \epsilon_{i(k)j(k)}$

	ten Hove et al. (2022)	Schönbrodt et al. (2022)	Correlation approach
Subject-level	$\frac{\sigma_S^2}{\sigma_S^2 + \sigma_\epsilon^2}$		$\mathrm{Cor}\left(Y_{i(k)j(k)}, Y_{i(k)j'(k)} \mid k, j, j' \right) = \frac{\sigma_S^2}{\sigma_S^2 + \sigma_\epsilon^2}$
R^S_{rel}		(not defined)	
R^S_{abs}	$\frac{\sigma_S^2}{\sigma_S^2 + \sigma_R^2 + \sigma_\epsilon^2}$ 	$\frac{\sigma_S^2}{\sigma_S^2 + \sigma_R^2 + \sigma_\epsilon^2}$ 	$\mathrm{Cor}\left(Y_{i(k)j(k)}, Y_{i(k)j'(k)} \mid k \right) = \frac{\sigma_S^2}{\sigma_S^2 + \sigma_R^2 + \sigma_\epsilon^2}$ 

Case 2: $Y_{i(k)j(k)} = \mu + C_k + S_{i(k)} + R_{j(k)} + \epsilon_{i(k)j(k)}$

	ten Hove et al. (2022)	Schönbrodt et al. (2022)	Correlation approach
Cluster-level			$\mathrm{Cor}\left(Y_{i(k)j(k)}, Y_{i(k)j'(k)} \mid i, j, j' \right) = \frac{\sigma_C^2}{\sigma_C^2 + \sigma_\epsilon^2}$
R_{rel}^{C}	Cannot be defined because it ignores all lower-level variances	(not defined)	
R_{abs}^{C}		$\frac{\sigma_C^2}{\sigma_C^2 + \sigma_S^2 + \sigma_R^2 + \sigma_\epsilon^2}$ 	$\mathrm{Cor}\left(Y_{i(k)j(k)}, Y_{i(k)j'(k)} \mid i \right) = \frac{\sigma_C^2}{\sigma_C^2 + \sigma_R^2 + \sigma_\epsilon^2}$ 

Case 3: $Y_{j(i(k))} = \mu + C_k + S_{i(k)} + \epsilon_{j(i(k))}$

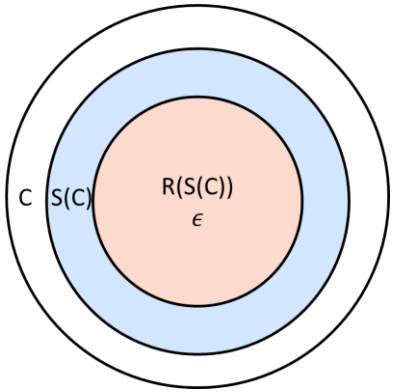
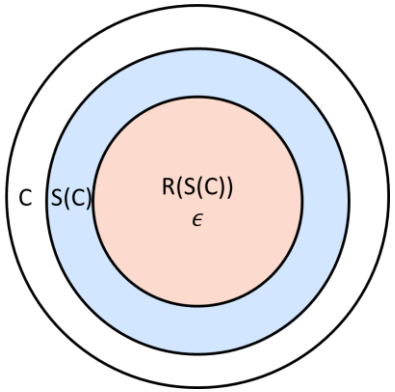
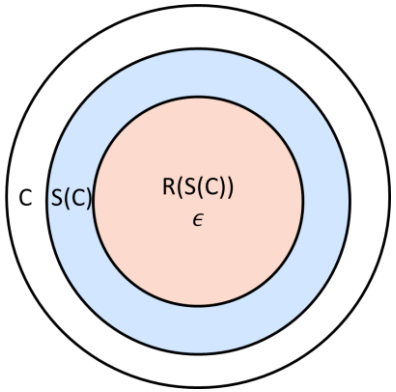
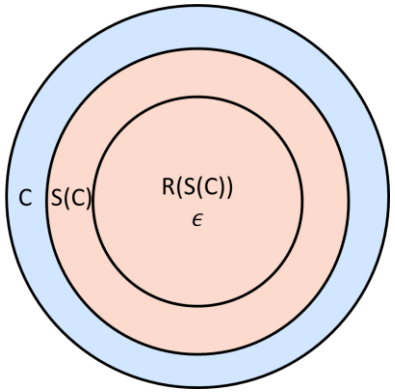
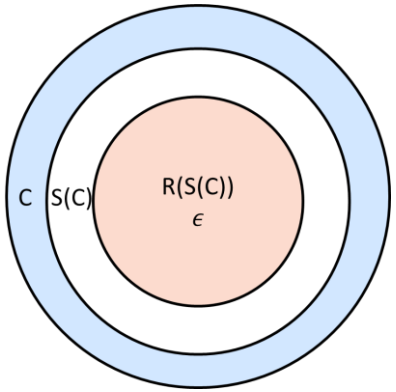
	ten Hove et al. (2022)	Schönbrodt et al. (2022)
Subject-level	$\frac{\sigma_S^2}{\sigma_S^2 + \sigma_\epsilon^2}$	$\frac{\sigma_S^2}{\sigma_S^2 + \sigma_\epsilon^2}$
R^S		
Cluster-level		$\frac{\sigma_C^2}{\sigma_C^2 + \sigma_S^2 + \sigma_\epsilon^2}$
R^C	Cannot be defined	

Correlation approach

$$\mathrm{Cor}\left(Y_{i(k)j}, Y_{i(k)j'} \mid k\right) = \frac{\sigma_S^2}{\sigma_S^2 + \sigma_\epsilon^2}$$

$$\mathrm{Cor}\left(Y_{i(k)j}, Y_{i(k)j'} \mid i\right) = \frac{\sigma_C^2}{\sigma_C^2 + \sigma_\epsilon^2}$$

Case 3: $Y_{j(i(k))} = \mu + C_k + S_{i(k)} + \epsilon_{j(i(k))}$

	ten Hove et al. (2022)	Schönbrodt et al. (2022)	Correlation approach
Subject-level	$\frac{\sigma_S^2}{\sigma_S^2 + \sigma_\epsilon^2}$	$\frac{\sigma_S^2}{\sigma_S^2 + \sigma_\epsilon^2}$	$\mathrm{Cor}\left(Y_{i(k)j}, Y_{i(k)j'} \mid k \right) = \frac{\sigma_S^2}{\sigma_S^2 + \sigma_\epsilon^2}$
R^S			
Cluster-level		$\frac{\sigma_C^2}{\sigma_C^2 + \sigma_S^2 + \sigma_\epsilon^2}$	$\mathrm{Cor}\left(Y_{i(k)j}, Y_{i(k)j'} \mid i \right) = \frac{\sigma_C^2}{\sigma_C^2 + \sigma_\epsilon^2}$
R^C	Cannot be defined		

Comparison

	ten Hove et al. (2022)	Schönbrodt et al. (2022)
Variance decomposition	Generalizability theory	
Reliability coefficient	In terms of ICCs	Extend from Shrout & Land's (2012) formulas which are based on the DC
Subject level	Do not contain any cluster-related (higher level) variance components (i.e., σ_C^2, σ_{CR}^2)	
Cluster level	Do not contain any subject-related variance (lower level) components (i.e., $\sigma_{S(C)}^2, \sigma_{S(C)R}^2$)	Contain subject-related variance components
Types	Separate *relative* and *absolute* reliability	Only discuss *absolute* reliability
Other characteristics	• Discuss subject-level and cluster-level reliability in three kinds of nested structure	• Discuss the roles of the *fixed effects* • Also build up the within-subject reliability under two different time scales

Intraclass Correlation Coefficient (ICC)

ICC is a widely used reliability coefficient in interrater reliability analyses (Bartko 1966; McGraw & Wang 1996; Shrout & Fleiss, 1979).

- Different forms of ICCs

	ICC	One-way $Y_{j(i)} = \mu + S_i + \epsilon_{j(i)}$	Two-way random effect $Y_{ij} = \mu + S_i + R_j + \epsilon_{ij}$	Two-way mixed effect $Y_{ij} = \mu + S_i + r_j + \epsilon_{ij}$
Single	Relative Consistency	$\frac{\sigma_S^2}{\sigma_S^2 + \sigma_\epsilon^2}$	$\frac{\sigma_S^2}{\sigma_S^2 + \sigma_\epsilon^2}$	$\frac{\sigma_S^2}{\sigma_S^2 + \sigma_\epsilon^2}$
	Absolute Agreement		$\frac{\sigma_S^2}{\sigma_S^2 + \sigma_R^2 + \sigma_\epsilon^2}$	$\frac{\sigma_S^2}{\sigma_S^2 + \theta_R^2 + \sigma_\epsilon^2}$

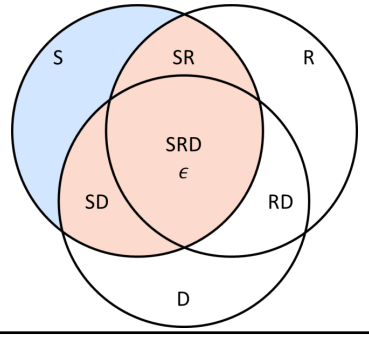
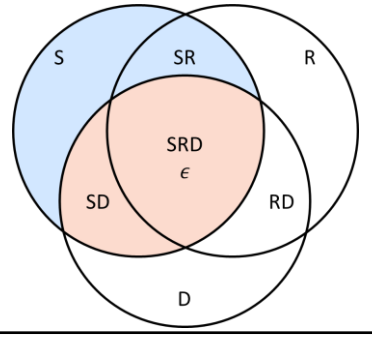
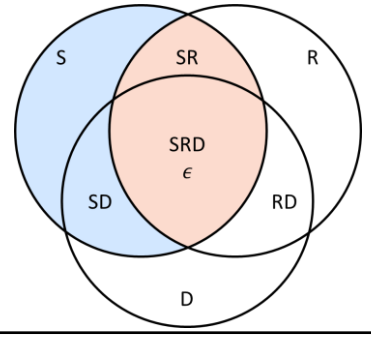
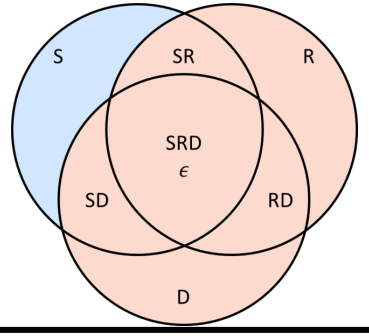
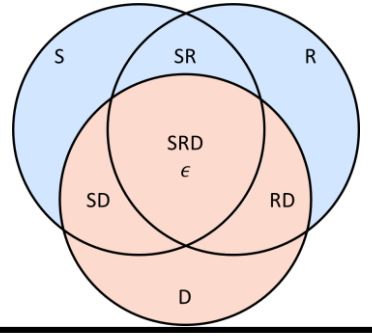
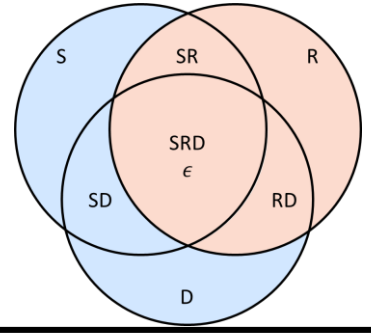
Intraclass Correlation Coefficient (ICC)

ICC is a widely used reliability coefficient in interrater reliability analyses (Bartko 1966; McGraw & Wang 1996; Shrout & Fleiss, 1979).

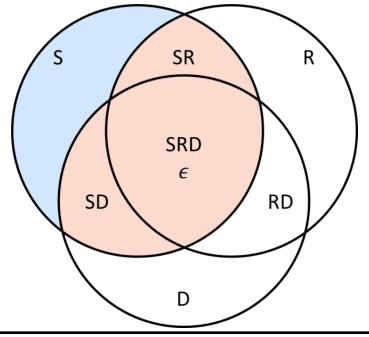
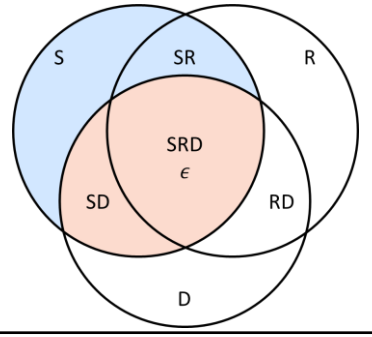
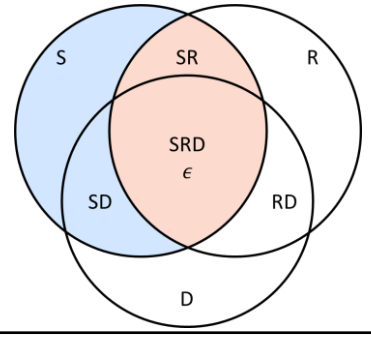
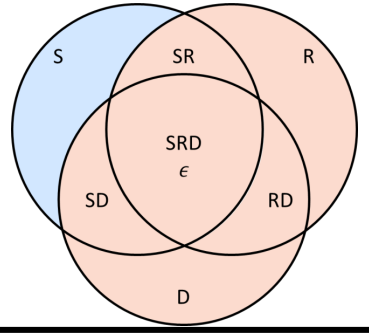
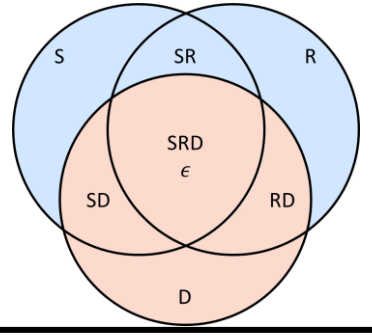
- Different forms of ICCs

	ICC	One-way $Y_{j(i)} = \mu + S_i + \epsilon_{j(i)}$	Two-way random effect $Y_{ij} = \mu + S_i + R_j + \epsilon_{ij}$	Two-way mixed effect $Y_{ij} = \mu + S_i + r_j + \epsilon_{ij}$
Single	Relative Consistency	$\frac{\sigma_S^2}{\sigma_S^2 + \sigma_\epsilon^2} = Cor(Y_{j(i)}, Y_{j'(i)})$	$\frac{\sigma_S^2}{\sigma_S^2 + \sigma_\epsilon^2} = Cor(Y_{ij}, Y_{ij'}\|j, j')$	$\frac{\sigma_S^2}{\sigma_S^2 + \sigma_\epsilon^2}$
	Absolute Agreement		$\frac{\sigma_S^2}{\sigma_S^2 + \sigma_R^2 + \sigma_\epsilon^2} = Cor(Y_{ij}, Y_{ij'})$	$\frac{\sigma_S^2}{\sigma_S^2 + \theta_R^2 + \sigma_\epsilon^2}$

GT vs. Correlation Approach

Reliability	Overall	Test-retest	Interrater
Relative Consistency	$\mathbf{GC} = \mathrm{Cor}(Y_{ijt}, Y_{ij't'} \mid j, j', t\ t') = \frac{\sigma_S^2}{\sigma_S^2 + \sigma_{SR}^2 + \sigma_{SD}^2 + \sigma_\epsilon^2}$ 	$\mathrm{Cor}(Y_{ijt}, Y_{ijt'} \mid j, t\ t') = \frac{\sigma_S^2 + \sigma_{SR}^2}{\sigma_S^2 + \sigma_{SR}^2 + \sigma_{SD}^2 + \sigma_\epsilon^2}$ 	$\mathrm{Cor}(Y_{ijt}, Y_{ij't} \mid j, j', t\ ) = \frac{\sigma_S^2 + \sigma_{SD}^2}{\sigma_S^2 + \sigma_{SR}^2 + \sigma_{SD}^2 + \sigma_\epsilon^2}$ 
Absolute Agreement	$\mathbf{DC} = \mathrm{Cor}(Y_{ijt}, Y_{ij't'}) = \frac{\sigma_S^2}{\sigma_S^2 + \sigma_R^2 + \sigma_D^2 + \sigma_{SR}^2 + \sigma_{SD}^2 + \sigma_{RD}^2 + \sigma_\epsilon^2}$ 	$\mathrm{Cor}(Y_{ijt}, Y_{ijt'}) = \frac{\sigma_S^2 + \sigma_R^2 + \sigma_{SR}^2}{\sigma_S^2 + \sigma_R^2 + \sigma_D^2 + \sigma_{SR}^2 + \sigma_{SD}^2 + \sigma_{RD}^2 + \sigma_\epsilon^2}$ 	$\mathrm{Cor}(Y_{ijt}, Y_{ij't}) = \frac{\sigma_S^2 + \sigma_D^2 + \sigma_{SD}^2}{\sigma_S^2 + \sigma_R^2 + \sigma_D^2 + \sigma_{SR}^2 + \sigma_{SD}^2 + \sigma_{RD}^2 + \sigma_\epsilon^2}$ 

GT vs. Correlation Approach

Reliability	Overall	Test-retest	Interrater
Relative Consistency	$\mathbf{GC} = \mathrm{Cor}(Y_{ijt}, Y_{ij't'} \mid j, j', t\ t') = \frac{\sigma_S^2}{\sigma_S^2 + \sigma_{SR}^2 + \sigma_{SD}^2 + \sigma_\epsilon^2}$ 	$\mathrm{Cor}(Y_{ijt}, Y_{ijt'} \mid j, t\ t') = \frac{\sigma_S^2 + \sigma_{SR}^2}{\sigma_S^2 + \sigma_{SR}^2 + \sigma_{SD}^2 + \sigma_\epsilon^2}$ 	$\mathrm{Cor}(Y_{ijt}, Y_{ij't} \mid j, j', t\ ) = \frac{\sigma_S^2 + \sigma_{SD}^2}{\sigma_S^2 + \sigma_{SR}^2 + \sigma_{SD}^2 + \sigma_\epsilon^2}$ 
Absolute Agreement	$\mathbf{DC} = \mathrm{Cor}(Y_{ijt}, Y_{ij't'}) = \frac{\sigma_S^2}{\sigma_S^2 + \sigma_R^2 + \sigma_D^2 + \sigma_{SR}^2 + \sigma_{SD}^2 + \sigma_{RD}^2 + \sigma_\epsilon^2}$ 	$\mathrm{Cor}(Y_{ijt}, Y_{ijt'}) = \frac{\sigma_S^2 + \sigma_R^2 + \sigma_{SR}^2}{\sigma_S^2 + \sigma_R^2 + \sigma_D^2 + \sigma_{SR}^2 + \sigma_{SD}^2 + \sigma_{RD}^2 + \sigma_\epsilon^2}$ 	$\mathrm{Cor}(Y_{ijt}, Y_{ij't}) = \frac{\sigma_S^2 + \sigma_D^2 + \sigma_{SD}^2}{\sigma_S^2 + \sigma_R^2 + \sigma_D^2 + \sigma_{SR}^2 + \sigma_{SD}^2 + \sigma_{RD}^2 + \sigma_\epsilon^2}$ 